

news reports and other programs subtitled by Chinese characters so that speakers of many different dialects can read and understand what is being said. But this does not mean that a billion Chinese speak Mandarin.

At the other end of the scale, the world map of languages shows several language families spoken by dwindling, often marginally located or isolated groups. The Indo-European languages of European colonizers surround the language families of Southeast Asia. Languages in the Austro-Asiatic language family survive in the interior of eastern India and in Cambodia and Laos. Languages in the Austronesian family are numerous and quite diverse, and many of the individual languages are spoken by fewer than 10 million people. Remoteness helps account for the remaining languages in the Amerindian language family. These languages remain strongest in areas of Middle America, the high Andes, and northern Canada.

If we look carefully at the map of world language families, some interesting questions arise. Consider, for example, the island of Madagascar off the East African coast. The primary languages people in Madagascar speak belong not to an African language family but to the Austronesian family, the languages of Southeast Asia and the Pacific Islands. Why is a language from this family spoken on an island so close to Africa? Anthropologists have found evidence of seafarers from the islands of Southeast Asia crossing the Indian Ocean to Madagascar. At the time, Africans had not sailed across the strait to Madagascar, so no African languages diffused to the island, preserving the Southeast Asian settlements and language for centuries. Later, Africans began to come to Madagascar, but by that time the language and culture of Southeast Asia had been well established.

Language Formation

In the process of classifying languages, linguists and linguistic geographers study relationships among languages, looking for similarities and differences within and among languages. One way to find and chart similarities among languages is to examine particular words, looking for cognates across languages. A **cognate** is a word that has the same linguistic derivation as another word. Take the case of Italian, Spanish, and French, all of which are members of the Romance language subfamily of the Indo-European language family because they are derived from Latin. The Latin word for milk, *lacte*, became *latte* in Italian, *leche* in Spanish, and *lait* in French; all are cognates. Also, the Latin for the number eight, *otto*, became *otto*, *ocho*, and *huit*, respectively. Even if linguists did not already know that Italian, Spanish, and French were languages rooted in Latin, they could deduce a connection among the languages through such cognates.

More than two centuries ago William Jones, an Englishman living in South Asia, undertook a study of Sanskrit, the language in which ancient Indian religious and literary texts were written. Jones discovered that the vocabulary and grammatical forms of Sanskrit bore a striking resemblance to the

ancient Greek and Latin he learned while in college. "No philologist [student of words] could examine all three," Jones wrote, "without believing them to have sprung from some common source, which, perhaps, no longer exists." His idea was a revolutionary notion in the 1700s.

During the nineteenth century Jakob Grimm, a scholar and a writer of fairy tales, suggested that cognates might prove the relationships between languages in a scientific manner. He explained that related languages have similar, but not identical, consonants. He believed these consonants would change over time in a predictable way. Hard consonants, such as the *v* and *t* in the German word *vater*, softened into *vader* (Dutch) and *father* (English). Using Grimm's theory that consonants became softer as time passed, linguists realized that consonants would become harder as they went "backwards" toward the original hearth and original language.

From Jones's notions and Grimm's ideas came the first major linguistic hypothesis, proposing the existence of an ancestral Indo-European language called **Proto-Indo-European**, which in turn gave rise to modern languages from Scandinavia to North Africa and from North America through parts of Asia to Australia. Studies of similarities across language families that occur less frequently suggest that Proto-Indo-European was, in turn, an outgrowth of an earlier **Proto-Eurasiatic** language. As the speakers of that language spread out and lost contact with one another some 15,000 years ago, variants of Proto-Eurasiatic emerged—a few of which gave rise to seven of the language families found in Eurasia today. Proto-Indo-European was one of these.

LOCATING THE HEARTH OF PROTO-INDO-EUROPEAN

German linguist August Schleicher was the first to compare the world's language families to the branches of a tree (Fig. 6.9). In the mid-nineteenth century, he suggested that new languages form through **language divergence**, which occurs when spatial interaction among speakers of a language breaks down and the language fragments first into dialects and then into discrete tongues. The process of language divergence has happened between Spanish and Portuguese and is now happening with Québécois French. Each new language becomes a new leaf on a tree, its branches leading back to the hearth, the trunk of the tree. Through **backward reconstruction** (tracking shifting consonants and cognates back in an effort to reconstruct elements of a prior common language), linguists and linguistic geographers can provide insight into how languages fit together and where the branches were once joined.

Finding the trunk of a language family is a daunting task, for reconstructing even a small branch of the language tree is complicated. Languages do not change solely through divergence (the splitting of branches); they also change through convergence and extinction. If peoples with different languages have consistent spatial interaction, **language convergence** can take place, collapsing two languages into

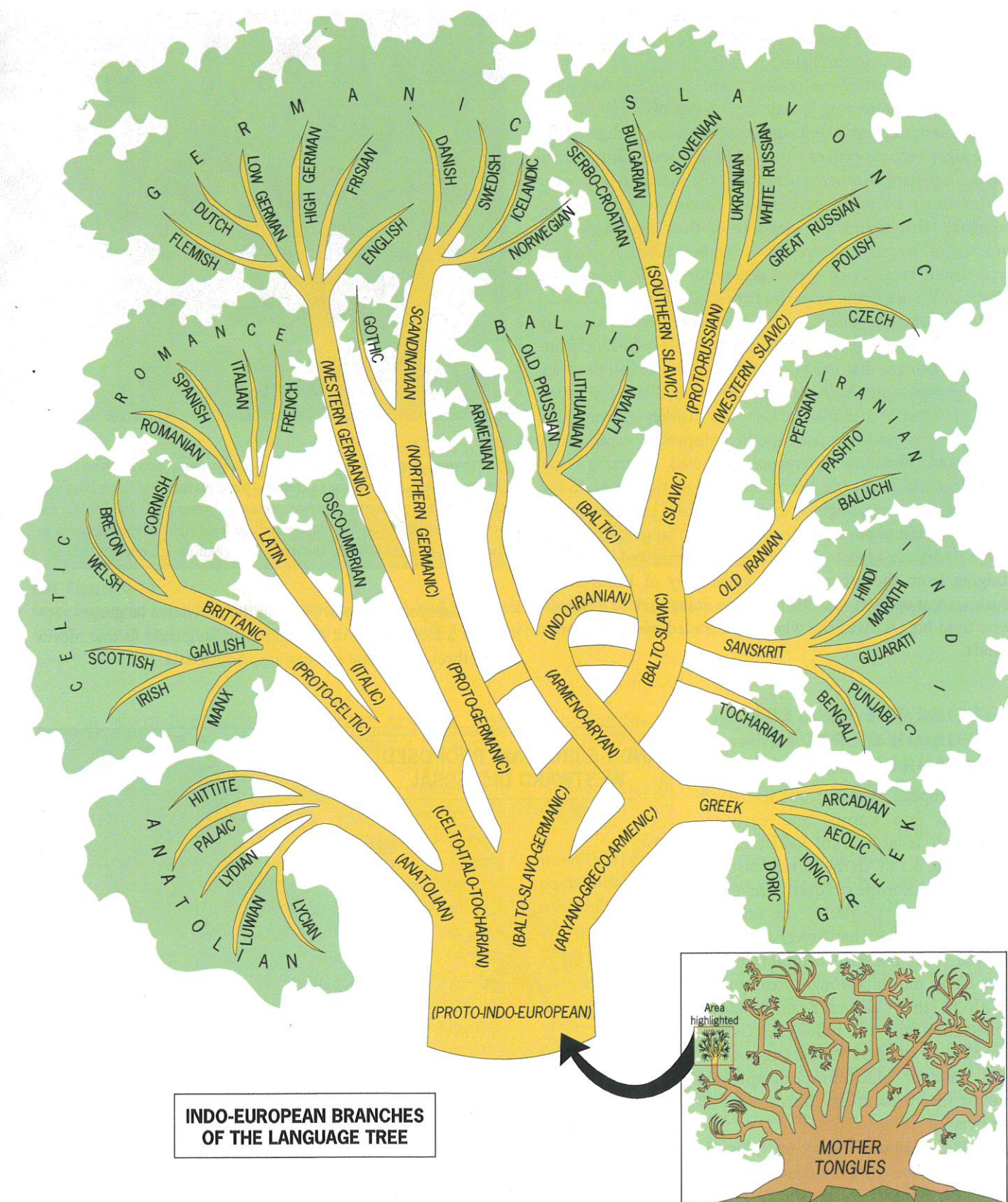


Figure 6.9

Indo-European Branches of the Language Tree. The inset shows all of the languages of the world in a massive tree. The Indo-European language family makes up just one part of that massive tree and is highlighted in this figure. Compare this figure to the map of the Indo-European languages in Europe (Figure 6.2) and notice how the language groups in Europe are related on this tree and where the languages on this tree are geographically located in Europe.